

Name: _____

Class _____



HSC Assessment 1 Term 4 2022

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 65 minutes
- Board approved calculators may be used.
- Write using black or blue pen
- All necessary working should be shown in all questions
- Write your student number at the top of every page of your answer sheets.
- NESAs Reference Sheet is provided

Total marks – 40

Attempt all questions

Section A – Answer on the Multiple Choice Answer Sheet

Section B - Start each question on a new sheet of paper.

NESA Reference Sheet is provided

This paper MUST NOT be removed from the examination room

Section A Multiple Choice (5 Marks)

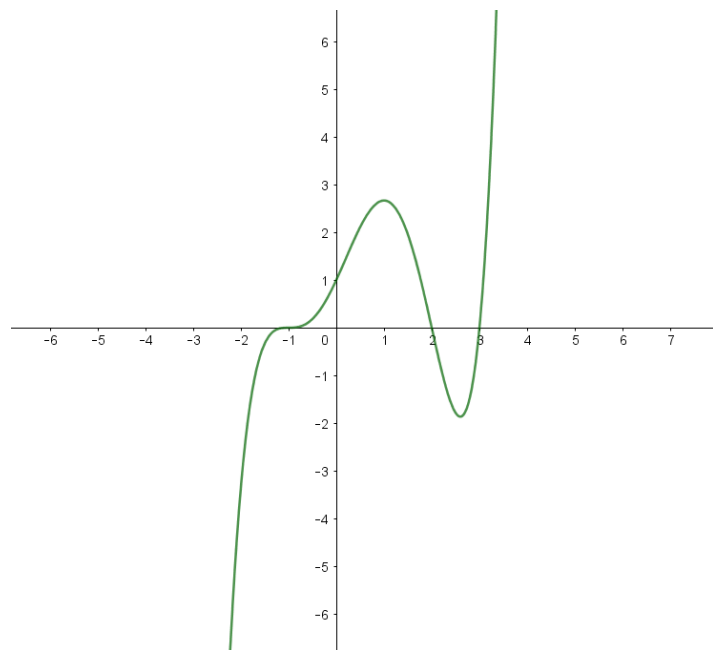
QUESTION 1

What is the remainder when $x^3 - 4x^2 + x + 1$ is divided by $2x + 1$?

- A. $\frac{5}{8}$
- B. $-\frac{5}{8}$
- C. -5
- D. -25

QUESTION 2

Which of the following options best represents the graph shown below?



- A. $y = (x + 1)^3(2 - x)(x - 3)$
- B. $y = (x + 1)^3(2 - x)(3 - x)$
- C. $y = \frac{1}{6}(x + 1)^3(2 - x)(x - 3)$
- D. $y = \frac{1}{6}(x + 1)^3(2 - x)(3 - x)$

QUESTION 3

The coefficient of x^2 in the expansion of $(3 - 5x)^7$ is

- A. 6075
- B. -6075
- C. 127575
- D. -127575

QUESTION 4

Which of the following step is not necessary to prove a statement to be true for all positive integer values n by mathematical induction?

- A. Prove that the statement is true for $n = 1$.
- B. Prove that if the statement is true for some case $n = k$, then the statement is also true for $n = k + 1$.
- C. Check that the statement is true for a very large value of n .
- D. None of the above.

QUESTION 5

Which of the following is the largest domain for x such that the equation $\cot\left(\frac{x}{2}\right) \sin x = 0$ has two solutions?

- A. $x \in [0, \pi]$
- B. $x \in [-\pi, \pi]$
- C. $x \in (-3\pi, 3\pi)$
- D. $x \in [0, 2\pi)$

SECTION B

QUESTION 6 Start a new page (5 marks)

Marks

- a) Expand $\left(\frac{2}{x} - 3x\right)^4$, simplify fully. 2
- b) Prove by mathematical induction that $7^{2n} + 7^n + 4$ is divisible by 6 for all positive integers n . 3

QUESTION 7 Start a new page (5 marks)

Marks

- a) The polynomial $P(x) = ax^4 + (a - b)x^3 - 2bx^2 + 12x + 8$ has a zero of multiplicity 2 at $x = 2$.
- i. By finding the values of a and b , show that remaining zeroes are -2 and $-\frac{1}{2}$. 3
- ii. Sketch the graph $y = P(x)$. 2
- DO NOT FIND THE COORDINATES OF TURNING POINTS.**

QUESTION 8 Start a new page (5 marks)

Marks

- a) In the polynomial $P(x) = (k^2 - 1)x^4 + 4x^3 - k^2x^2 + x + 3 - 3k$, solve for k such that the leading coefficient is equal to the constant term. 2
- b) Find the coefficient of x^4 in the expansion of $\left(1 - \frac{2}{x}\right)^4 (x + 3)^5$. 3

QUESTION 9 Start a new page (5 marks)

Marks

- a)
- i. Express $\cos 2\theta - \sqrt{3} \sin 2\theta$ in the form $R \cos(2\theta + \alpha)$ where $R > 0$ and $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Show all necessary working. 2
- ii. Hence solve for θ in the equation $\cos 2\theta = \sqrt{3} \sin 2\theta + 1$, where $\theta \in [-\pi, \pi]$. 2
- iii. For what value(s) of c will the equation $\cos 2\theta = \sqrt{3} \sin 2\theta + c$ have exactly 4 solutions when $\theta \in [-\pi, \pi]$? 1

QUESTION 10 Start a new page (4 marks)**Marks**

a)

i. Show that $4k^3 + 12k^2 + 11k + 3 = (k + 1)(4(k + 1)^2 - 1)$.

1

ii. Hence, prove by mathematical induction that

3

$$1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(4n^2 - 1)}{3}$$

for all positive integers n .**QUESTION 11 Start a new page (6 marks)****Marks**

a) Prove that $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ by considering two different ways of selecting k objects from a group of n .

3

DO NOT PROVE THIS ALGEBRAICALLY, THIS PROOF IS TO BE COMBINATORIAL.

b)

i. By making the substitution $t = \tan \frac{\theta}{2}$, show that $\sqrt{\frac{1+\cos \theta}{1-\cos \theta}} = \cot \frac{\theta}{2}$ for $\theta \in (0, \pi)$.**2**ii. Hence or otherwise, find the exact value of $\cot \frac{\pi}{8}$ in the form $\sqrt{\frac{\sqrt{a}+b}{\sqrt{c}-d}}$.**1****QUESTION 12 Start a new page (5 marks)****Marks**a) $P(x)$ is a polynomial of degree 4. It is known that $P(x) \geq x$ for $x \in \mathbb{R}$ and that**2**

$$P(1) = 1$$

$$P(2) = 4$$

$$P(3) = 3$$

Find $P(4)$.

b) Solve:

$$\sum_{r=0}^{2n} \binom{2n}{r} \cos^{4n-2r} \theta \sin^{2r} \theta = 16^n \cos^{4n} \theta$$

3for $\theta \in [0, 2\pi]$.**End of examination**

MATHEMATICS EXT 1: Question 6

Suggested Solutions	Marks	Marker's Comments
a, $\left(\frac{2}{x} - 3x\right)^4$ $= \left(\frac{2}{x}\right)^4 - 4\left(\frac{2}{x}\right)^3(3x) + 6\left(\frac{2}{x}\right)^2(3x)^2 - 4\left(\frac{2}{x}\right)(3x)^3 + (3x)^4$ $= \frac{16}{x^4} - \frac{96}{x^1} + 216 - 216x^2 + 81x^4$	① ①	
b, $n=1 \rightarrow 7^{2 \cdot 1} + 7^1 + 4 = 49 + 7 + 4$ $= 60$ $= 6 \times 10$ which is div. by 6 $\therefore$ true for $n=1$ Assume that for some $k \geq 1$, $7^{2k} + 7^k + 4 = 6M$ where $M \in \mathbb{Z}$ $\Leftrightarrow 7^{2k} = 6M - 4 - 7^k, M \in \mathbb{Z}$ (*) Now, $7^{2(k+1)} + 7^{k+1} + 4$ $= 7^{2k} \cdot 7^2 + 7^k \cdot 7 + 4$ $= (6M - 4 - 7^k) \cdot 7^2 + 7^k \cdot 7 + 4$, by (*) $= 49(6M) - 4(49) - 49(7^k) + 7(7^k) + 4$ $= 6(49M) - 4 \cdot 49 - 48 \cdot 7^k$ $= 6(49M) - 4(6 \times 7) - (6 \times 8)7^k$ $= 6[49M - 28 + 8 \cdot 7^k]$ (*) $= 6N$, where $N = 49M - 28 + 8 \cdot 7^k$ is an integer since $\mathbb{Z}$ is closed under addition, subtraction and multiplication. $\therefore 7^{2(k+1)} + 7^{k+1} + 4$ is div. by 6 if $7^{2k} + 7^k + 4$ is div. by 6. $\therefore$ Statement is true for integers $n \geq 1$ by Mathematical Induction.	① ①	Base Case Correct use of assumption (reference assumption) Not marked down here, but this should be included. Correctly completing proof.

Question 7

a)

i. $P(x) = ax^4 + (a - b)x^3 - 2bx^2 + 12x + 8$

$$P'(x) = 4ax^3 + 3(a - b)x^2 - 4bx + 12 \quad \text{1 mark for differentiating}$$

Zero of multiplicity 2 at $x = 2$.

$$P(2) = 0 \rightarrow 16a + 8(a - b) - 8b + 24 + 8 = 0$$

$$24a - 16b = -32$$

$$3a - 2b = -4 \text{ — Equation 1}$$

$$P'(2) = 0 \rightarrow 32a + 12(a - b) - 8b + 12 = 0$$

$$44a - 20b = -12$$

$$11a - 5b = -3 \text{ — Equation 2}$$

$$(5 \times \text{Equation 1}) - (2 \times \text{Equation 2}) \rightarrow 7a = 14$$

$$a = 2$$

$$\text{Sub } a = 2 \text{ into Equation 1} \rightarrow 6 - 2b = -4$$

$$b = 5 \quad \text{1 mark for solving for } a \text{ and } b$$

$$\therefore P(x) = 2x^4 - 3x^3 - 10x^2 + 12x + 8$$

$$P(-2) = 2(-2)^4 - 3(-2)^3 - 10(-2)^2 + 12(-2) + 8$$

$$= 32 + 24 - 40 - 24 + 8$$

$$= 0$$

$$P\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^4 - 3\left(-\frac{1}{2}\right)^3 - 10\left(-\frac{1}{2}\right)^2 + 12\left(-\frac{1}{2}\right) + 8$$

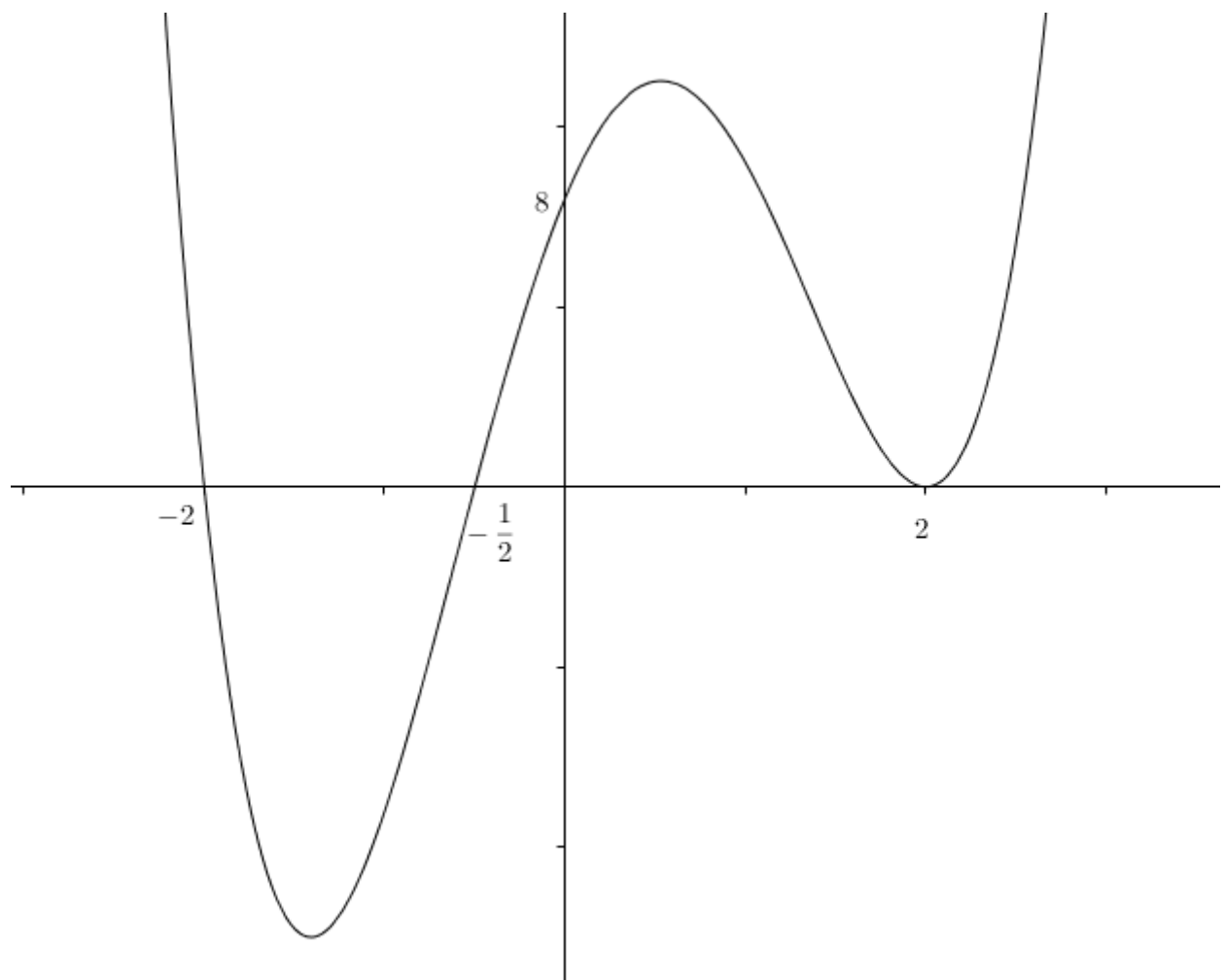
$$= \frac{1}{8} + \frac{3}{8} - \frac{10}{4} - 6 + 8$$

$$= 0$$

$$\therefore x = -2 \text{ and } x = -\frac{1}{2} \text{ are zeroes.} \quad \text{1 mark for answer showing all steps}$$

Note: If you chose to solve for the remaining zeroes by using sums and products of roots then, then you must solve for BOTH α and β respectively to get -2 and $-\frac{1}{2}$, and not just conclude your proof complete just because you get two possible values for α .

ii.



1 mark for correct shape

1 mark for all other details such as intercepts and scale

(However you can't get a mark for just drawing any graph with the correct intercepts)

8. a)

$$(k^2 - 1) = 3 - 3k$$

$$k^2 + 3k - 4 = 0$$

$$(k+4)(k-1) = 0$$

$$k = -4 \text{ or } 1$$

but if $k=1$, then the leading term is $4x^3$.

$\therefore k \neq 1$, $\therefore k = -4$ only

$$b) (1-2x^{-1})^4 (x+3)^5 = \sum_{r=0}^4 \binom{4}{r} (1)^r (-2x^{-1})^{4-r} \cdot \sum_{k=0}^5 \binom{5}{k} x^k (3)^{5-k}$$

$$\therefore \text{general term is } \binom{4}{r} \binom{5}{k} (-2)^{4-r} (3)^{5-k} x^{r-4+k}$$

$$\text{term for } x^4 \Rightarrow r-4+k = 4$$

$$r+k = 8$$

$$\therefore r=3, k=5$$

$$r=4, k=4$$

$$\therefore \text{coefficient is given by } \left[\binom{4}{3} \binom{5}{5} (-2)^1 (3)^0 + \binom{4}{4} \binom{5}{4} (-2)^0 (3)^1 \right]$$

$$= -8 + 15$$

$$= 7$$

Question 9

i) Let $R \cos(2\theta + \alpha) = \cos 2\theta - \sqrt{3} \sin 2\theta$
 $\therefore R \cos 2\theta \cos \alpha - R \sin 2\theta \sin \alpha = \cos 2\theta - \sqrt{3} \sin 2\theta$

$$\therefore R \cos \alpha = 1 \quad \text{--- Equation 1}$$

$$R \sin \alpha = \sqrt{3} \quad \text{--- Equation 2}$$

Since $R > 0$,

$$\therefore \alpha \in \left(0, \frac{\pi}{2}\right)$$

$$\text{Equation 2} \div \text{Equation 1} \rightarrow \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}, \left(\alpha \in \left(0, \frac{\pi}{2}\right)\right)$$

Sub $\alpha = \frac{\pi}{3}$ into Equation 1:

$$R \cos \frac{\pi}{3} = 1$$

$$R = 1 \div \frac{1}{2}$$

$$= 2$$

$$\therefore \cos 2\theta - \sqrt{3} \sin 2\theta = 2 \cos \left(2\theta + \frac{\pi}{3}\right)$$

(1st mark for solving one of R or α , 2nd mark for the other variable AND answering the question)

ii) $\cos 2\theta = \sqrt{3} \sin 2\theta + 1$
 $\cos 2\theta - \sqrt{3} \sin 2\theta = 1$
 $2 \cos \left(2\theta + \frac{\pi}{3}\right) = 1$
 $\cos \left(2\theta + \frac{\pi}{3}\right) = \frac{1}{2}$
 $2\theta + \frac{\pi}{3} = \pm \frac{\pi}{3}, \pm \frac{5\pi}{3}, \pm \frac{7\pi}{3} \dots$
 $2\theta = 0, -\frac{2\pi}{3}, \frac{4\pi}{3}, -2\pi, 2\pi$
 $\theta = 0, -\frac{\pi}{3}, \frac{2\pi}{3}, -\pi, \pi$

If $\theta \in [-\pi, \pi]$, then $2\theta \in [-2\pi, 2\pi]$

(1st mark for solving some correct values of θ , 2nd mark for all correct solutions of θ)

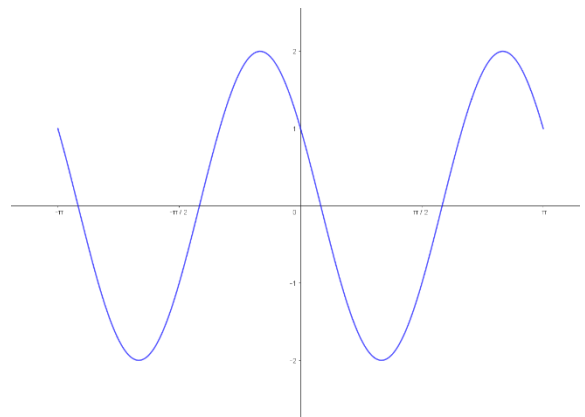
iii) $\cos 2\theta = \sqrt{3} \sin 2\theta + c$
 $\cos 2\theta - \sqrt{3} \sin 2\theta = c$
 $2 \cos \left(2\theta + \frac{\pi}{3}\right) = c$

As the period of the function is π , there will always be 4 solutions when $x \in [-\pi, \pi]$ except for when

$$c = \pm 2, 1.$$

$$\therefore c \in (-2, 1) \cup (1, 2) \text{ or}$$

$$-2 < c < 2, c \neq 1.$$



1 mark for correct answer

Maths Ext 1 Question 10.

$$\begin{aligned}
 \text{a) i) RHS} &= (k+1)(4(k+1)^2-1) \\
 &= (k+1)(4k^2+8k+4-1) \\
 &= (k+1)(4k^2+8k+3) \\
 &= 4k^3+8k^2+3k+4k^2+8k+3 \\
 &= 4k^3+12k^2+11k+3 \\
 &= \text{LHS}
 \end{aligned}$$

OR

$$\begin{aligned}
 \text{LHS} &= 4k^3+12k^2+11k+3 \\
 &= 4k^3+4k^2+8k^2+8k+3k+3 \\
 &= 4k^2(k+1)+8k(k+1)+3(k+1) \\
 &= (k+1)(4k^2+8k+3) \\
 &= (k+1)(4k^2+8k+4-1) \\
 &= (k+1)(4(k^2+2k+1)-1) \\
 &= (k+1)(4(k+1)^2-1)
 \end{aligned}$$

a) ii) For $n=1$

$$\begin{aligned}
 \text{LHS} &= 1^2 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{RHS} &= \frac{(1)(4 \times 1^2-1)}{3} \\
 &= 1
 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$, true for $n=1$.

Assume statement true for $n=k$
 i.e. $1^2+3^2+5^2+\dots+(2k-1)^2 = \frac{k(4k^2-1)}{3}$

Prove statement true for $n=k+1$

$$\begin{aligned}
 &1^2+3^2+5^2+\dots+(2k-1)^2+(2(k+1)-1)^2 \\
 &= \frac{k(4k^2-1)}{3} + (2k+1)^2 \quad \text{By assumption}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{k(4k^2-1)+3(2k+1)^2}{3} \\
 &= \frac{4k^3-k+12k^2+12k+3}{3}
 \end{aligned}$$

$$= \frac{4k^3+12k^2+11k+3}{3}$$

$$= \frac{(k+1)(4(k+1)^2-1)}{3} \quad \text{from part i).}$$

$\therefore$ Statement is true for $n=k+1$, given it is true for $n=k$.

$\therefore$ Since statement is true for $n=1$, from above it must be true for $n=1+1=2$, $2+1=3$ and so on for all positive integers n .

①

For 1 mark long division is time wasting.

①

You need to show some substitution.

$$\text{LHS} = 1$$

$$\text{RHS} = 1$$

is insufficient.

State assumption clearly.

①

make it clear where you used assumption.

show steps clearly

①

"Hence..." means you must use part i) and clearly state where you used it.

QUESTION 11 Start a new page (6 marks)

Marks

- a) Prove that $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ by considering two different ways of selecting k objects from a group of n .

3

DO NOT PROVE THIS ALGEBRAICALLY, THIS PROOF IS TO BE COMBINATORIAL.

b)

- i. By making the substitution $t = \tan \frac{\theta}{2}$, show that $\frac{1+\cos \theta}{1-\cos \theta} = \cot^2 \frac{\theta}{2}$ for $\theta \in (0, \pi)$.

2

- ii. Hence or otherwise, find the exact value of $\cot \frac{\pi}{8}$ in the form $\sqrt{\frac{a+b}{c-d}}$.

1

11. a) To choose k objects from n :
1st way: nC_k (By definition)

2nd way: Consider 1 object from the list, if it is included, then there are $n-1$ objects left to choose $k-1$, $\therefore {}^{n-1}C_{k-1}$.
If it is not included, then we need to choose k from $n-1$, which is ${}^{n-1}C_k$.
 $\therefore$ Total is ${}^{n-1}C_{k-1} + {}^{n-1}C_k$

$$\therefore {}^nC_k = {}^{n-1}C_{k-1} + {}^{n-1}C_k$$

1 mark for recognising the case where a specific object is **NOT** included and obtaining the correct expression from it.

1 mark for doing the same with the case where a specific object **MUST** be included

1 mark for connecting that the sum of the 2 cases must be the same as n choose k .

b)

$$\text{Given } t = \tan \theta/2$$

$$\text{So } \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\therefore \text{L.H.S} = \sqrt{\frac{1 + \frac{1-t^2}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}}}$$

$$= \sqrt{\frac{1+t^2 + 1-t^2}{1+t^2 - 1+t^2}}$$

$$= \sqrt{\frac{2}{2t^2}}$$

(1)

$$= \frac{1}{t} \text{ as } t > 0 \text{ for } \theta \in (0, \pi)$$

$$= \cot \theta/2$$

$$= \text{R.H.S}$$

2nd mark

(ii)

from above

$$\cot \pi/8 = \sqrt{\frac{1 + \cos \pi/4}{1 - \cos \pi/4}}$$

$$= \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}} = \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}}$$

(1)

MATHEMATICS Extension 1: Question 12

Suggested Solutions

Marks

Marker's Comments

a) $P(x)$ - polynomial degree 4.

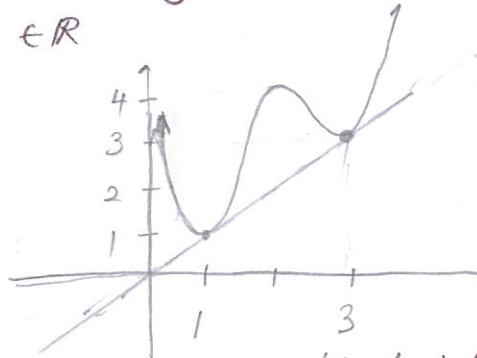
$$P(x) \geq x, \quad x \in \mathbb{R}$$

$$P(1) = 1$$

$$P(2) = 4$$

$$P(3) = 3$$

Find $P(4)$.



Method 1

Diagram reveals that if $(P(x) \geq x) \wedge P(1) = 1$ and $P(3) = 3$, then $P(x)$ must be tangential to $y=x$ at $x=1 \wedge x=3$. i.e. double roots to the eqn $P(x) - x = 0$ or $P(x) = x$

$$\text{Let } P(x) = ax^4 + bx^3 + cx^2 + dx + e$$

$\therefore ax^4 + bx^3 + cx^2 + (d-1)x + e = 0$ has 4 roots

$$\boxed{1, 1, 3, 3}$$

$$x^4 - (\alpha + \beta + \gamma + \delta)x^3 + (\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)x^2 - (\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta)x + \alpha\beta\gamma\delta = 0$$

using sum and product of roots identity

$$-\frac{b}{a} = 8 \quad (\text{sum of roots}) \quad (1)$$

$$b = -8a \quad \text{--- i}$$

$$\frac{c}{a} = (1+3+3) + (3+3) + (3 \times 3)$$

$$c = 22a \quad \text{--- ii}$$

$$-\frac{(d-1)}{a} = 3+3+9+9 \quad (\text{sum of products of each triplet of roots}) \quad (3)$$

$$d = 1 - 24a \quad \text{--- iii}$$

$$\frac{e}{a} = 9 \quad (\text{product of roots}) \quad (4)$$

$$e = 9a \quad \text{--- iv}$$

- understanding double roots and forming an eqn.

- finding the polynomial eqn.

(1) - evaluating $P(4)$

Poorly done!

Students who attempted SE, didn't get anywhere because they didn't apply the identity rules.

- students didn't understand $P(x) \geq x$ thus the mistake

MATHEMATICS: Question.....

Suggested Solutions	Marks Awarded	Marker's Comments
if $P(2) = 4$, then $16a + 8b + 4c + 2d + e = 4 \quad \text{--- v}$ sub i, ii, iii & v into v.. $16a + 8(-8a) + 4(22a) + 2(1 - 24a) + 9a = 4$ $a = 2$ $b = -16$ $c = 44$ $d = -47$ $e = 18$ $\therefore P(x) = 2x^4 - 16x^3 + 44x^2 - 47x + 18$ $\therefore P(4) = 22 \quad \text{--- (1)}$ Method 2. Let $Q(x) = P(x) - x$ and $P(1) = 1 \Rightarrow Q(1) = 0$ $P(3) = 3 \Rightarrow Q(3) = 0$ $x = 1$ & 3 are zeros of $Q(x)$ with m2, since $Q(x) \geq 0$. $Q(x) = A(x-1)^2(x-3)^2$ $P(2) = 4 \Rightarrow Q(2) + 2 = 4 \quad \leftarrow Q(2) = P(2) - 2$ $A(2-1)^2(2-3)^2 = 2 \therefore A = 2$ $Q(x) = 2(x-1)^2(x-3)^2$ $P(4) = Q(4) + 4$ $= 2(4-1)^2(4-3)^2 + 4$ $= 22 \quad \text{--- (1)}$		1 mark for 3 correct value No marks awarded if answer was correct and no working out (1) to show relevant working out to find value of A.

MATHEMATICS Extension 2: Question

Method 1	Suggested Solutions	Marks	Marker's Comments
b) Solve $\sum_{r=0}^{2n} \binom{2n}{r} \cos^{4n-2r} \theta \sin^{2r} \theta = 16^n \cos^{4n} \theta$ for $\theta \in [0, 2\pi]$ $\sum_{r=0}^{2n} \binom{2n}{r} (\cos^2 \theta)^{2n-r} (\sin^2 \theta)^r = 16^n \cos^{4n} \theta$ $(\cos^2 \theta + \sin^2 \theta)^{2n} = 16^n \cos^{4n} \theta$ $1^{2n} = 16^n \cos^{4n} \theta$ $\frac{1^{2n}}{16^n} = \cos^{4n} \theta$ $\cos^{4n} \theta = \left(\frac{1}{16}\right)^n$ $\cos^4 \theta = \frac{1}{16}$ $\cos \theta = \pm \frac{1}{2}$ $\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$	or $* \frac{1^{2n}}{4^n} = 2^{4n} \cos^{4n} \theta$ $\frac{1}{4^n} = \cos^{4n} \theta$ $\left[\begin{array}{c c} 120^\circ & 60^\circ \\ \hline 240^\circ & 300^\circ \end{array} \right]$ acute angle = 60 $\therefore \theta = 60^\circ$		poorly done!
Method 2 LHS $\binom{2n}{0} \cos^{4n} \theta + \binom{2n}{1} \cos^{4n-2} \theta \sin^2 \theta + \binom{2n}{2} \cos^{4n-4} \theta \sin^4 \theta + \dots + \binom{2n}{2n-1} \cos^2 \theta \sin^{4n-2} \theta + \binom{2n}{2n} \sin^{4n} \theta$ $= \cos^{4n} \theta + \binom{2n}{1} \cos^{4n-2} \theta \sin^2 \theta + \binom{2n}{2} \cos^{4n-4} \theta \sin^4 \theta + \dots + \binom{2n}{2n-1} \cos^2 \theta \sin^{4n-2} \theta + \binom{2n}{2n} \sin^{4n} \theta$ $= 1 + \binom{2n}{1} \frac{\sin^2 \theta}{\cos^2 \theta} + \binom{2n}{2} \frac{\sin^4 \theta}{\cos^4 \theta} + \dots + \binom{2n}{2n-1} \frac{\sin^{4n-2} \theta}{\cos^{4n-2} \theta} + \binom{2n}{2n} \frac{\sin^{4n} \theta}{\cos^{4n} \theta}$ $= 1 + \binom{2n}{1} \tan^2 \theta + \binom{2n}{2} \tan^4 \theta + \dots + \binom{2n}{2n-1} \tan^{4n-2} \theta + \binom{2n}{2n} \tan^{4n} \theta$ $= (1 + \tan^2 \theta)^{2n}$	$\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$		1 mark for each pair of correct answer. ① - for working

MATHEMATICS Extension 2: Question

Suggested Solutions

Marks

Marker's Comments

$$(1 + \tan^2 \theta)^{2n} = \frac{16^n \cos^{4n} \theta}{\cos^{4n} \theta} \quad (\text{as done on LHS})$$

$$(1 + \tan^2 \theta)^{2n} = 16^n$$

$$1 + \tan^2 \theta = 16^{n/2n}$$

$$\tan^2 \theta = \sqrt{16} - 1$$

$$= 3$$

$$\tan \theta = \pm \sqrt{3}$$

$$\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3} \quad \text{--- } \textcircled{2}$$

1 mark for each correct pair of answers